Vishwakarma Institute of Technology **Issue 01 : Rev No. 00 : Dt. 01/08/22**

Title : Project Registration & Progress Review **FF No 180**

Department: Computer Engineering	**Academic Year:** 2025-26
Semester: 4	**Group No:** SEDA CS B8
Project Title: Ayurvedic Decision Support System for Dosha Detection from Clinical Vital Signs and Rule-Based Herbal Medication Recommendation	
Project Area: AI and IoT integration in Healthcare and Medication System	

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Project approved

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Project Synopsis:

INTRODUCTION

Ayurveda, the ancient Indian system of medicine dating back over 5,000 years, emphasizes personalized healthcare through the concept of **Prakriti** (constitutional typology) and **Tridosha** (three biological humors: Vata, Pitta, and Kapha). According to Ayurvedic principles, an individual's Prakriti is determined at conception and remains constant throughout life, governing physiological, psychological, and pathological tendencies. Traditional Prakriti assessment relies on subjective questionnaires and expert pulse diagnosis (Nadi Pariksha), which suffer from inter-observer variability and lack standardization. With the convergence of artificial intelligence (AI), Internet of Things (IoT), and biomedical signal processing, there exists a transformative opportunity to establish objective, reproducible, and scalable frameworks for Prakriti assessment using modern vital sign measurements.

Recent advances in wearable sensor technology and machine learning have enabled continuous, non-invasive monitoring of physiological parameters such as heart rate (HR), blood oxygen saturation (SpO_2), and body temperature. Research demonstrates that these vital signs correlate with Ayurvedic Dosha characteristics: Vata exhibits irregul ar heart rate patterns and fluctuating oxygen levels, Pitta shows elevated heart rates and consistently high body temperatures, while Kapha presents lower heart rates and stable but reduced oxygen saturation. The labeled data will train machine learning models capable of real-time Prakriti classification, subsequently deployed on low-cost IoT hardware (MAX30100 pulse oximeter, DS18B20 temperature sensor, Arduino/ESP32) for point-of-care diagnostics. This work contributes to the emerging field of Ayurinformatics and personalized digital health.

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PROBLEM STATEMENT

Despite growing interest in integrating traditional medicine with modern technology, Man critical challenges hinder the development of objective Ayurvedic Prakriti assessment systems. Traditional Prakriti assessment methods (questionnaires, pulse diagnosis) suffer from subjectivity, lack of reproducibility, and dependence on expert availability, making them unsuitable for scalable digital health applications. Third, while research papers theoretically map vital signs to Dosha characteristics (e.g., irregular HR → Vata), there is no validated IoT-based real-time implementation demonstrating feasibility in resource-constrained settings using affordable wearable sensors. This project addresses these gaps by creating a novel PhysioNet-Dosha dataset through literature-derived rule-based labeling, training machine learning models to achieve >80%

classification accuracy, and deploying the trained model on MAX30100 and DS18B20 sensors for real-time Dosha prediction, thereby enabling accessible, objective, and continuous Prakriti monitoring in both clinical and home healthcare environments.

OBJECTIVES

The primary objectives of this research project are:

- **Dataset Creation and Labelling:**
  Develop a novel PhysioNet–Dosha labeled dataset by applying rule-based weak supervision derived from AI–Tridosha literature to PhysioNet Challenge 2019 vital signs data (heart rate, SpO_2, and temperature). This process will generate more than 40,000 labeled instances categorized as Vata-, Pitta-, or Kapha-dominant.

- **Feature Engineering:**
  Extract temporal features from 5-minute vital sign windows, including mean, standard deviation, variability, and trend patterns, to capture Dosha-specific physiological characteristics.

- **Machine Learning Model Development and Evaluation:**
  Train and evaluate supervised learning models such as Random Forest and XGBoost using the labeled dataset, targeting a classification accuracy greater than 80 percent. Model interpretability will be ensured through feature importance analysis using SHAP (SHapley Additive exPlanations).

- **Future IoT Hardware Integration:**
  Design and implement a low-cost wearable system using the MAX30100 pulse oximeter for heart rate and SpO_2 measurement and the DS18B20 temperature sensor, interfaced with an Arduino or ESP32 microcontroller. The system will acquire real-time physiological data at 100 millisecond intervals.

- **Herbal Recommendation System:** Design and implement a rule-based herbal recommendation engine that maps detected Dosha imbalances to evidence-based Ayurvedic herbs (e.g., Ashwagandha for Vata, Shatavari for Pitta, Holy Basil for Kapha), integrating classical Ayurvedic texts with modern phytochemical research.

LITERATURE REVIEW

Paper Title	Authors	Year	Key Results	Methodology	Limitations Addressed
AI-Driven Tridosha Analysis for Precision in Ayurvedic Healthcare	Deshmukh et al.	2024	Established HR-SpO_2-Temp mapping to Vata/Pitta/Kapha; 85%+ accuracy with neural networks	Pulse oximeters, thermometers, pacemaker data analyzed via ML	No public dataset, no herb recommendations; we create PhysioNet-Dosha labels + herbal system
Artificial Intelligence for Objective Prakriti Assessment	Kumar & Singh	2025	ML models achieved >85% agreement with expert assessments using phenotypic data	Questionnaire-based ML (SVM, RF) with NLP on text responses	Lacks continuous vitals and treatment; we use time-series IoT + herbs
PhysioNet Challenge 2019: Early Prediction of Sepsis	Reyna et al.	2019	40,336 ICU records with HR, SpO_2, Temp, BP sampled hourly for sepsis prediction	Time-series feature engineering + ensemble methods	Sepsis-focused; we repurpose for Dosha classification
Comparative Analysis of ML for Prakriti Identification	Sharma et al.	2025	XGBoost with one-hot encoding achieved 91.2% accuracy on questionnaire data (807 subjects)	Hyperparameter tuning + ensemble stacking on categorical data	Small dataset, no IoT; we scale to 40K+ samples with sensors
Nadi Pariksha with Technological Aid: A Review	Fatangare & Joshi	2024	Impedance plethysmography (IPG) pulse signals correlate with Tridosha; standardization needed	Signal processing of PPG waveforms for pulse quality assessment	Lab-based; we deploy on low-cost IoT sensors + herbs
Establishing Scientific Validity of Tridosha	Hankey	2012	Systems biology approach validates Dosha theory through genomic and physiological correlations	Factor analysis + Ayurgenomics mapping	Theoretical; we provide computational implementation + treatment

PROPOSED METHODOLOGY

1. Data Acquisition and Preprocessing

- **Dataset Selection:** The PhysioNet Challenge 2019 dataset containing 40,336 ICU patient records is used. From multiple vitals, only heart rate, oxygen saturation, and body temperature are selected due to their relevance to Tridosha analysis.
- **Data Cleaning:** Missing values are handled using forward-fill interpolation for up to two hours, while records with excessive missing data are removed. Physiologically unrealistic values are filtered using clinically accepted ranges.

2. Rule-Based Labelling

- **Vata Classification:** Samples showing high variability in heart rate, oxygen saturation, and temperature are labelled as Vata-dominant, reflecting unstable physiological behavior.
- **Pitta Classification:** Samples with consistently high heart rate, high oxygen saturation, and elevated body temperature is labelled as Pitta-dominant.
- **Kapha Classification:** Samples characterized by low heart rate, stable oxygen saturation, and lower body temperature is labelled as Kapha-dominant.

3. Machine Learning Model Training and Evaluation

- **Supervised Models:** Random Forest and XGBoost models are trained using stratified cross-validation, with XGBoost serving as the primary model.
- **Performance Evaluation:** Accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC are used to assess classification performance, along with SHAP-based interpretability.

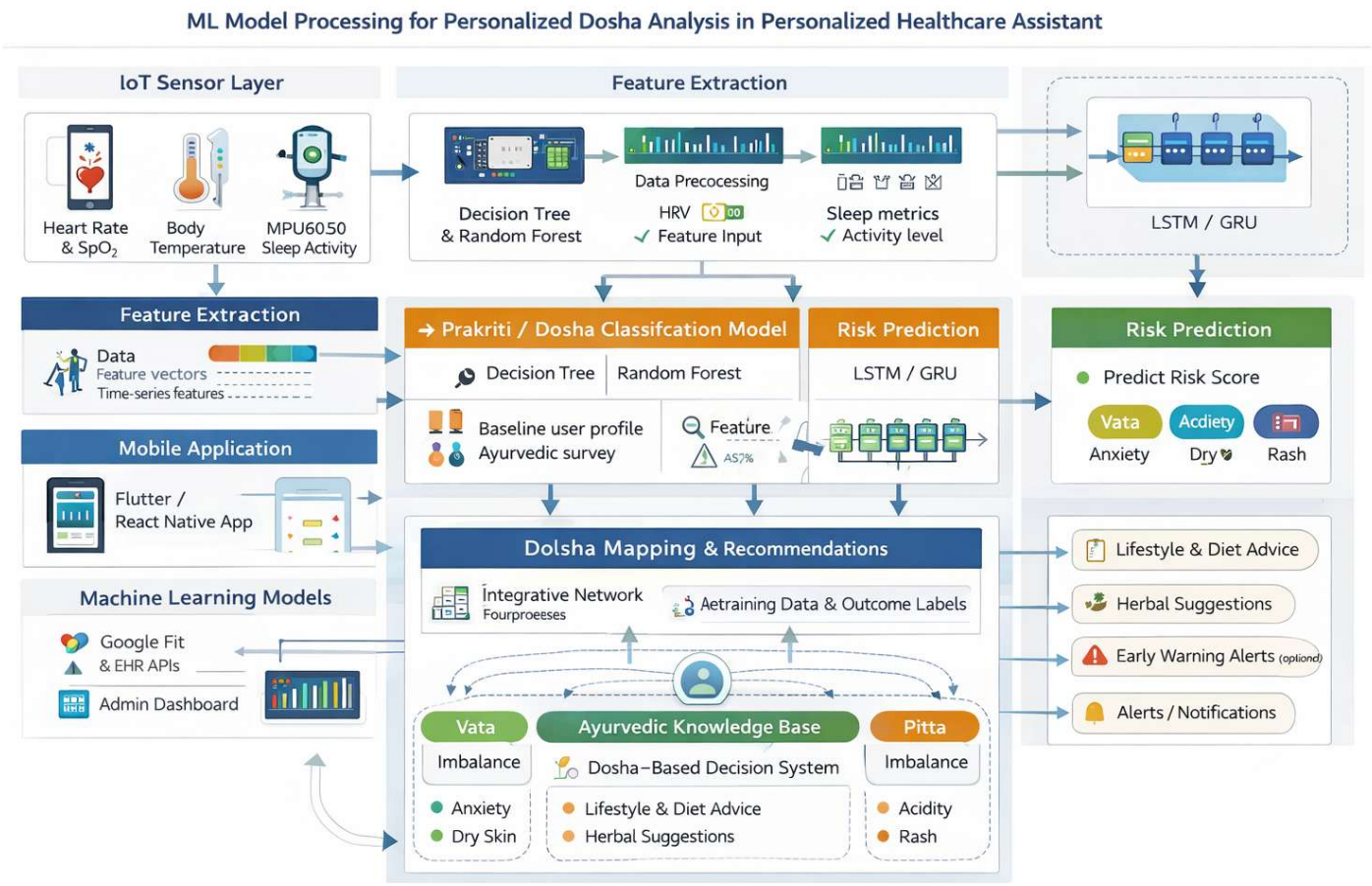
4. Herbal Recommendation Engine

We curate a scientifically-validated herbal database from classical Ayurvedic texts (Charaka Samhita, Ashtanga Hridaya) and modern research, containing 20+ herbs mapped to Dosha effects

5. Future IoT hardware System Design

- **MAX30100:** Integrated pulse oximeter and heart rate sensor (I2C interface, 100Hz sampling)
- **DS18B20:** Digital temperature sensor (1-Wire protocol, ±0.5°C accuracy)
- **ESP32:** Microcontroller with Wi-Fi/Bluetooth, 240MHz dual-core, 4MB flash

ARCHITECTURE DIAGRAM



Group No.			
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	**Group No.:**	**Date:**
Progress Review Report		
Signature of Guide:		

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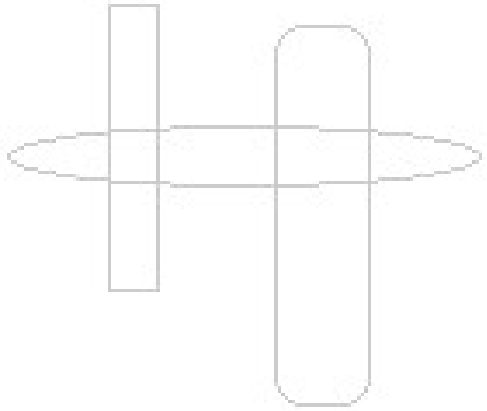
Review No.: 2	**Group No.:**	**Date:**

Progress Review Report

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